

Module 03 - Advanced SQL

Exercise 1: Ranking and Window Functions

Step 1: Create Database

```
CREATE DATABASE CompanyDB;  
USE CompanyDB;
```

Step 2: Create Table

```
CREATE TABLE Employees (  
EmployeeID INT PRIMARY KEY,  
EmployeeName VARCHAR(50),  
Department VARCHAR(30),  
Salary INT  
);
```

Step 3: Insert Data

```
INSERT INTO Employees VALUES  
(101,'Ritesh','IT',60000),  
(102,'Rahul','IT',75000),  
(103,'Aman','HR',50000),  
(104,'Priya','HR',65000),  
(105,'Neha','Finance',80000),  
(106,'Ankit','Finance',70000);
```

ROW_NUMBER()

```
SELECT *, ROW_NUMBER() OVER(ORDER BY Salary DESC) AS RowNum FROM Employees;
```

RANK()

```
SELECT *, RANK() OVER(ORDER BY Salary DESC) AS RankNo FROM Employees;
```

DENSE_RANK()

```
SELECT *, DENSE_RANK() OVER(ORDER BY Salary DESC) AS DenseRank FROM Employees;
```

PARTITION BY

```
SELECT *, ROW_NUMBER() OVER(PARTITION BY Department ORDER BY Salary DESC) AS  
DeptRank FROM Employees;
```

Learning Outcome

Learned ROW_NUMBER(), RANK(), DENSE_RANK(), PARTITION BY and SQL Window Functions.